Arjun Biddanda, Ph.D.

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Research Interests

Statistical Genetics, Population Genomics, Reproductive Genetics, Ancient DNA

Professional Experience

2026 - Present — Associate Research Scientist · Department of Biology · Johns Hopkins University
Advisor: Rajiv C. McCoy

2023 - 2026 — Postdoctoral Fellow · Department of Biology · Johns Hopkins University
Advisor: Rajiv C. McCoy

2021 - 2023 — Computational Scientist · Genomics & Data Science · 54gene

2021 — Postdoctoral Research Associate · Department of Statistics · University of Oxford
Advisor: Pier Francesco Palamara

Education

2015 - 2020 — PhD · Human Genetics · University of Chicago
Dissertation: *Investigating the spatio-temporal structure of human genetic diversity*
Advisor: John Novembre

2015 - 2020 — M.Sc · Human Genetics · University of Chicago

2011 - 2015 — B.S · Computer Science (*Cum Laude*) · Cornell University
Advisor: Alon Keinan

Funding

2024 - 2026 — Lalor Foundation Fellowship (PI: Arjun Biddanda)
Clarifying the origins and fitness effects of human segmental aneuploidies ($110,000)

Honors & Awards

2024 — Best Poster Award
MD-GEM Training Grant Meeting, Johns Hopkins University

2021 — Postdoctoral Associate
Corpus Christi College, University of Oxford

2020 — Presidential Membership
Genetics Society of America

2019 — Reviewer's Choice Abstract
American Society of Human Genetics

2015 - 2018 — NIH Genetics and Regulation Training Grant Fellow
University of Chicago

Publications

* - *equal contribution*; † - *corresponding author*

Preprints

2026 O. S. Harringmeyer, **A. Biddanda**, R. C. McCoy, and J. M. Akey. Gene flux shapes diversity and evolution of the ancient 17q21.31 inversion polymorphism. *bioRxiv*.

Peer-reviewed Publications

2026 B. Jeffrey, Y. Wong, ..., **A. Biddanda**, ..., P. L. Ralph, and J. Kelleher. Population-scale ancestral recombination graphs with tskit 1.0. *Nature Methods (accepted)*.

2026 Y. Zhang*,†, **A. Biddanda***,† , S. Johnson, C. O'Dushlaine, and P. Moorjani†. Recovering signatures of archaic introgression using ancestral recombination graphs. *Science*, 0(1):eaef8874.

2026 A. Das, **A. Biddanda**, R. C. McCoy, and M. C. Schatz. Assembling unmapped reads reveals hidden variation in South Asian genomes. *Nature Communications*, 17(1):5095.

2026 S. A. Carioscia*, **A. Biddanda*** , M. R. Starostik, X. Tang, E. R. Hoffman, Z. P. Demko, and R. C. McCoy. Common variation in meiosis genes shapes human recombination phenotypes and aneuploidy risk. *Nature*, 651(1):1–8.

2025 J. A. Urban, ..., **A. Biddanda**, ..., N. Rai, and M. Raghavan. Population histories of the Indigenous Adivasi and Sinhalese from Sri Lanka using whole genomes. *Current Biology*, 35(11):2554–2566.

2025 M. C. Steiner, D. P. Rice, **A. Biddanda**, M. K. Ianni-Ravn, C. Porras, and J. Novembre. Study design and the sampling of deleterious rare variants in biobank-scale datasets. *PNAS*, 122(23):e2425196122.

2025 D. Yoo, A. Rhie, ..., **A. Biddanda**, ..., A. M. Phillippy, and E. E. Eichler. Complete sequencing of ape genomes. *Nature*, 641:1–18.

2024 K. D. Makova, B. D. Pickett, ..., **A. Biddanda**, ..., E. E. Eichler, and A. M. Phillippy. The complete sequence and comparative analysis of ape sex chromosomes. *Nature*, 630(8016):401–411.

2024 **A. Biddanda*** , E. Bandyopadhyay*, C. de la Fuente Castro*, D. Witonsky, ..., C. D. Kodira, A. P. Naren, M. Sikdar, N. Rai, and M. Raghavan. Distinct positions of genetic and oral histories: Perspectives from India. *Human Genetics and Genomics Advances*, 5(3):e100305.

2024 D. J. Taylor, S. B. Chhetri, M. G. Tassia, **A. Biddanda**, S. M. Yan, G. L. Wojcik, A. Battle, and R. C. McCoy. Sources of gene expression variation in a globally diverse human cohort. *Nature*, 632(8032):122–130.

2023 E. Joshi, **A. Biddanda**, J. Popoola, A. Yakubu, O. Osakewe, D. Attipoe, 54gene Team, NCD-GHS Consortium, E. Dogbo, B. Salako, O. Nash, O. Salako, O. Oyedele, G. Eze-Echesi, S. Fatumo, A. Ene-Obong, and C. O'Dushlaine. Whole-genome sequencing across 449 samples spanning 47 ethnolinguistic groups provides insights into genetic diversity in Nigeria. *Cell Genomics*, 3(9):e100378.

2023 B. C. Zhang, **A. Biddanda**, A. F. Gunnarson, F. Cooper, and P. F. Palamara. Biobank-scale inference of ancestral recombination graphs enables genealogy-based mixed model association of complex traits. *Nature Genetics*, 55(5):768–776.

2023 M. E. Lauterbur, M. I. A. Cavassim, A. L. Gladstein, ..., **A. Biddanda**, ..., P. L. Ralph, D. R. Schrider, and I. Gronau. Expanding the stdpopsim species catalog, and lessons

learned for realistic genome simulations. *eLife*, 12:e84874.

2022 C. Washington III, M. Dapas, **A. Biddanda**, K. M. Magnaye, ..., C. G. McKennan, and C. Ober. African-specific alleles modify risk for asthma at the 17q12-q21 locus in African Americans. *Genome Medicine*, 14(1):112.

2022 **A. Biddanda**, M. Steinrücken, and J. Novembre. Properties of two-locus genealogies and linkage disequilibrium in temporally structured samples. *Genetics*, 221(1):eiyac038.

2020 **A. Biddanda**, D. P. Rice, and J. Novembre. Geographic patterns of human allele frequency variation: a variant-centric perspective. *eLife*, page e60107.

2018 C. W. K. Chiang, J. H. Marcus, C. Sidore, **A. Biddanda**, H. Al-Asadi, M. Zoledziewska, M. Pitzalis, F. Busonero, A. Maschio, G. Pistis, M. Steri, A. Angius, K. E. Lohmueller, G. R. Abecasis, D. Schlessinger, F. Cucca, and J. Novembre. Genomic history of the Sardinian population. *Nature Genetics*, 50(10):1426–1434.

2018 P. de Barros Damgaard, R. Martiniano, J. Kamm, J. V. Moreno-Mayar, ..., **A. Biddanda**, ..., M. Sikora, A. K. Outram, R. Durbin, and E. Willerslev. The first horse herders and the impact of early Bronze Age steppe expansions into Asia. *Science*, 360(6396):eaar7711.

2016 Y. Y. Waldman*, **A. Biddanda*** , N. R. Davidson, P. Billing-Ross, M. Dubrovsky, C. L. Campbell, C. Oddoux, E. Friedman, G. Atzmon, E. Halperin, H. Ostrer, and A. Keinan. The genetics of Bene Israel from India reveals both substantial Jewish and Indian ancestry. *PLoS One*, 11(3):e0152056.

2016 Y. Y. Waldman, **A. Biddanda**, M. Dubrovsky, C. L. Campbell, C. Oddoux, E. Friedman, G. Atzmon, E. Halperin, H. Ostrer, and A. Keinan. The genetic history of Cochin Jews from India. *Human Genetics*, (135):1–17.

2015 F. Gao*, D. Chang*, **A. Biddanda*** , L. Ma, Y. Guo, Z. Zhou, and A. Keinan. XWAS: a software toolset for genetic data analysis and association studies of the X chromosome. *Journal of Heredity*, 106(5):666–671.

Review Articles

2026 **A. Biddanda**, S. A. Carioscia, E. R. Hoffmann, and R. C. McCoy. The spectrum of de novo lethal mutations in human prenatal development. *Nature Reviews Genetics (in revision)*.

Presentations

Invited Presentations

2026 Department of Genetic Medicine, Johns Hopkins University.
Evolutionary Genomics across Human Populations and Families.

2026 Evolutionary Genetics Seminar Series, University at Buffalo.
Common variation in meiosis genes shapes human recombination phenotypes and aneuploidy risk.

2025 Center for Theoretical and Evolutionary Genomics, UC Berkeley.
Common variation in meiosis genes shapes human recombination phenotypes and aneuploidy risk.

2024 Statistical Genetics Working Group, Johns Hopkins University.
Genetic architecture and fitness costs of meiotic recombination across 138,495 in vitro fertilized embryos.

2023 Department of Biology, Johns Hopkins University.
The spatio-temporal structure of human genetic diversity.

2021 Department of Statistics, University of Oxford.
Theoretical models for haplotype diversity with time-stratified sampling.

2021 Population Genomics Group, UC Berkeley.

Geographic patterns of human allele frequency variation: a variant-centric perspective.

Contributed Presentations

2026 American Society of Human Genetics, Montreal, Canada.
Genetic architecture and fitness costs of chromosomal abnormalities from 99,765 products of conception (*Platform Talk*).

2026 Modeling and Theory in Population Biology, ICERM, Providence, RI.
Recovering signatures of archaic introgression using ancestral recombination graphs.

2026 CAG-SURF PhD Course, University of Copenhagen (Online).
Human Fertility: Genome Stability Across Human Conceptions.

2025 Origins of Aneuploidy Research Consortium Meeting, Leuven, Belgium.
Common variation in meiosis genes shapes human recombination phenotypes and aneuploidy risk.

2024 American Society of Human Genetics, Denver, CO.
Genetic architecture of meiotic recombination across 99,495 in vitro fertilized embryos. (Plenary Talk; ‡RCM presented due to parental leave).

2023 American Society of Human Genetics, Washington, DC.
Genetic architecture of meiotic recombination across 69,223 in vitro fertilized embryos (Platform Talk).

2019 Midwest Population Genetics Meeting, Chicago, IL.
Linkage Disequilibrium in Ancient DNA: Theory and Applications.

Poster Presentations

2025 American Society of Human Genetics. Boston, MA.
Evidence of transmission distortion across 139,850 blastocyst-stage human embryos.

2024 Biology of Genomes, Cold Spring Harbor Laboratory, NY.
Genetic architecture of meiotic recombination across 99,132 in vitro fertilized embryos.

2024 The Allied Genetics Conference. National Harbor, MD.
Two-Locus Genealogies under Isolation-by-Distance and Time.

2022 American Society of Human Genetics. Los Angeles, CA.
Recovering signatures of ghost admixture using genealogies.

2021 Probabilistic Models in Genomics. (Online).
Two-locus genealogies and linkage disequilibrium in temporally stratified samples.

2019 American Society of Human Genetics. Houston, TX.
Linkage Disequilibrium in Ancient DNA: Theory and Applications.

2018 American Society of Human Genetics. San Diego, CA.
Inference and visualization of the geographic distribution for variant sets.

Teaching Experience

Winter 2023 Guest Speaker, *Johns Hopkins University*
Population Genetics Simulation and Visualization

Winter 2019 Guest Lecturer, *University of Chicago*
Computing Skills for Biologists

Winter 2018 Teaching Assistant, *University of Chicago*
HG 486: Fundamentals of Computational Biology

Summer 2017 Course Assistant, *University of Chicago*
Marine Biological Laboratory - Quantitative Approaches to Biology Bootcamp

Spring 2017	Teaching Assistant, *University of Chicago* HG 469: Human Variation and Disease
2016 - 2018	Tutor, *University of Chicago* Introduction to Statistics for Geneticists
2016 - 2022	Instructor, *Software Carpentry*
2013 - 2015	Teaching Assistant, *Cornell University* CS 3110 : Functional Programming and Data Structures

Academic Service

2025 - 2026	Communications Liason *JHU Biology Postdoc Association*
2023	Teaching Certificate *Johns Hopkins University Teaching Institute*
2022	ASHG Session Chair / Organizer *Demographic history, natural selection, and disease risk in diverse global biobanks*
2021 - 2024	Peer Review Training Program *Genetics Society of America*
Summer 2020	Co-Organizer *Genetics and Society Reading Group (Departmental Group)*
2019-2020	Novembre - He - Stephens (NHS) Meeting Coordinator *University of Chicago*
2019 - Present	*Ad Hoc* Peer Review *Nature, Nature Genetics, Genetics, eLife, Molecular Biology and Evolution, Genome Biology and Evolution, Scientific Reports, Frontiers in Genetics, Molecular Ecology Resources, BMC Genomics, Bioinformatics Advances*

Outreach / Science Communication

2024 - Present	Skype a Scientist
2023 - 2026	CareerSpring Advisor

Mentorship

Graduate

Winter 2025	Carsten Hoeke (*CMDB Rotation Student at Johns Hopkins*) *Project: Evaluating genetic effects on crossover interference strength*
Fall 2025	Brooke Weingard (*CMDB Rotation Student at Johns Hopkins*) *Project: Estimating relatedness across multiple reproductive testing datasets*
Spring 2025	Olivia Choi (*CMDB Rotation Student at Johns Hopkins*) *Project: Genetic influences on euploid and aneuploid pregnancy losses*
Spring 2024	Maya Mastronado (*CMDB Rotation Student at Johns Hopkins*) *Project: Quantifying centromere-haplotype effects on aneuploidy risk*

Undergraduate

2025 - Present	Angela Yang (*Johns Hopkins Undergraduate*) *Project: Spatial statistical models of embryo development and rebiopsy bias*

2024	Ameena K. Beg (*Johns Hopkins Undergraduate*) *Project: Evolutionary history of a human mitotic aneuploidy risk locus*
2022 - 2024	Axel Zagal-Norman (*UNAM Undergraduate Internship*) *Project: Visualization of Linkage Disequilibrium across multiple populations* *Current: PhD Student in Genetics at U. Chicago*
Summer 2023	Emma M. Smith (*NSF REU Trainee at Johns Hopkins*) *Project: Genomic basis of dosage imbalance in human embryonic aneuploidy* *Current: MSc Student in Genomic Medicine at Oxford*
2020 - 2021	Achyutha Menon (*U. Chicago Undergraduate*) *Project: Storage and visualization of multi-population Linkage Disequilibrium* Sponsored by a College Summer Research Fellowship *Current: MS Student in Computer Science at UCSD*

Society Memberships

2025 - Present	Origins of Aneuploidy Research Consortium (OARC)
2018 - Present	American Society of Human Genetics (ASHG)
2021 - Present	Genetics Society of America (GSA)
2024 - Present	Society for Modeling and Theory in Population Biology (SMTPB)
2021 - 2025	Society for Molecular Biology and Evolution (SMBE)